

Decision-Aware Validation of Consumer Credit Risk Models under Distributional Stress: Calibration, Utility Optimization, Subgroup Robustness, and Bootstrap Inference

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Abstract

Consumer credit models are commonly reported as classification systems, yet credit approval is fundamentally an economic decision made under uncertainty. A model that improves accuracy can still generate poor portfolio outcomes if its probability estimates are miscalibrated, if the operating threshold is inconsistent with financial utility, or if errors concentrate within borrower subgroups during stressed conditions. This study develops a decision-aware validation framework for consumer credit modeling using the UCI Default of Credit Card Clients dataset. Logistic Regression, Random Forest, and Gradient Boosting are evaluated through predictive discrimination, probability calibration, expected financial utility, false approval risk, false rejection risk, subgroup robustness, simulated distributional stress, bootstrap confidence intervals, and paired model comparison. The contribution is methodological rather than merely predictive: the analysis reframes default modeling from a narrow supervised-learning exercise into a decision-risk validation problem. The proposed framework demonstrates how probability quality, threshold policy, asymmetric economic consequences, subgroup heterogeneity, stress sensitivity, and statistical uncertainty should be jointly examined before a credit model is interpreted as reliable for lending decisions.

Keywords: consumer credit risk; default prediction; calibration; expected utility; stress testing; bootstrap inference; subgroup robustness; decision-risk modeling.

1. Introduction

Credit risk prediction is a high-impact domain in applied statistical learning because model outputs directly influence financial exposure, borrower access to credit, and institutional risk management. In consumer lending, a lender typically estimates the probability that an applicant or account will default, and that estimate becomes an input to approval, denial, pricing, credit-line assignment, reserve planning, or manual review. Because these actions carry economic consequences, the validity of a credit model cannot be established solely by conventional classification accuracy.

A central limitation of prediction-only evaluation is that it treats errors as symmetric. In lending, however, false approvals and false rejections are not equivalent. A false approval may create default loss and collection cost, whereas a false rejection may create lost interest income, reduced portfolio growth, and possible access-to-credit concerns. The relevant research question is therefore not only whether a model predicts default labels correctly, but whether the model supports decisions that remain economically coherent under uncertainty.

This study is motivated by the gap between predictive performance and decision quality. Standard benchmark exercises often ask which model achieves the highest accuracy, F1 score, or AUC. A more rigorous credit-risk question is how model probabilities behave when transformed into lending decisions under explicit financial assumptions, subgroup conditions, and distributional stress. This distinction matters because credit portfolios are not stationary. Borrower repayment behavior can deteriorate when utilization rises, payment delays increase, unemployment shocks occur, inflation reduces disposable income, or macroeconomic stress changes the distribution of risk.

The empirical setting uses the UCI Default of Credit Card Clients dataset. Three model families are examined: Logistic Regression as an interpretable linear baseline, Random Forest as a bagged nonlinear ensemble, and Gradient Boosting as a sequential nonlinear ensemble. The objective is not to declare a simplistic leaderboard winner. Instead, the analysis develops a multidimensional validation framework linking model probabilities to threshold policy, expected utility, calibration, subgroup behavior, stress robustness, and uncertainty quantification.

The study addresses five research questions. First, how do standard credit models differ when evaluated by financial utility rather than accuracy alone? Second, are predicted default probabilities sufficiently calibrated to support thresholding and risk-based decision-making? Third, how sensitive are model decisions and portfolio outcomes to simulated deterioration in borrower conditions? Fourth, are model errors and utility outcomes stable across borrower subgroups? Fifth, are observed differences in expected utility statistically meaningful under paired bootstrap resampling?

2. Research Contributions and Hypotheses

This paper makes four methodological contributions. First, it reframes consumer credit modeling as a decision-risk validation problem rather than a label-prediction problem. Second, it integrates discrimination, calibration, thresholding, expected utility, subgroup diagnostics, stress testing, and bootstrap inference in one reproducible pipeline. Third, it treats model comparison as conditional on explicit economic assumptions rather than as a fixed algorithmic ranking. Fourth, it provides a research presentation structure suitable for explaining why credit-model validation requires probability quality, policy design, and uncertainty estimation.

The empirical analysis is organized around the following hypotheses.

- **H1:** Nonlinear ensemble models produce a more favorable utility-risk tradeoff than a purely linear baseline when borrower risk patterns contain nonlinear interactions.
- **H2:** Calibration quality materially affects threshold-based lending decisions because the numerical default probability is used as a decision input.
- **H3:** Simulated distributional stress reduces portfolio utility and increases false approval risk relative to the baseline condition.
- **H4:** Borrowers with prior payment delay exhibit larger degradation under stress than the full evaluation population.
- **H5:** Paired bootstrap intervals provide a more defensible basis for model comparison than single point estimates.

3. Research Motivation and Related Concepts

Credit scoring has a long methodological history in statistics, econometrics, and financial risk management. Logistic Regression remains a common baseline because its coefficients are interpretable, its probabilistic form is easy to explain, and its assumptions can be documented for governance purposes. In regulated or audit-sensitive settings, transparency is

not merely convenient; it supports reproducibility, challenge, validation, and stakeholder communication.

Nevertheless, borrower behavior is often nonlinear. Repayment history, statement balances, utilization pressure, credit limits, and payment amounts may interact in ways that a linear model cannot fully represent. Tree-based ensembles such as Random Forest and Gradient Boosting can capture nonlinear thresholds and higher-order interactions without requiring every relationship to be manually specified. This flexibility can improve discrimination, but it also introduces additional validation needs, especially around calibration, stability, and interpretability.

Discrimination metrics remain useful but incomplete. AUC summarizes the model’s ability to rank defaulters above non-defaulters across thresholds. It does not determine which threshold should be used, whether predicted probabilities match observed default frequencies, or whether a particular approval policy is economically sensible. Similarly, accuracy can be misleading when the target class is imbalanced or when different errors have different costs.

Probability calibration is therefore central to this study. Calibration evaluates whether predicted probabilities correspond to empirical frequencies. For example, among borrowers assigned an estimated default probability of 20 percent, approximately 20 percent should default in a well-calibrated system. A model can rank borrowers effectively while still producing probabilities that are systematically too high or too low. For credit policy, this distinction is material because pricing, reserve planning, and approval thresholds often depend on the numerical probability, not only the rank order.

Stress robustness is equally important. Models trained on historical data may perform well under ordinary conditions but degrade when borrower characteristics shift. This paper treats stress testing as a validation exercise rather than a macroeconomic forecast. The stress scenarios are controlled perturbations applied to risk-relevant variables. Their purpose is to examine relative model sensitivity when repayment behavior and utilization conditions worsen.

Finally, uncertainty quantification strengthens model comparison. A single train-test split provides an estimate, not a permanent truth. Bootstrap resampling allows the analysis to estimate uncertainty around average utility, expected loss, calibration error, and model differences. Paired bootstrap testing is especially appropriate because competing models are evaluated on the same resampled observations, making the distribution of within-sample differences the object of inference.

4. Data Source and Variable Structure

The empirical analysis is based on the UCI Default of Credit Card Clients dataset, a public borrower-level dataset containing 30,000 observations and a binary target indicating whether a client defaulted on payment in the following month. Although the dataset is not a contemporary production portfolio, it is suitable for reproducible methodological research because it contains repayment history, bill amounts, payment amounts, credit limits, and demographic variables.

The variables can be organized into conceptually meaningful categories. Credit capacity is represented by credit limit. Repayment status variables capture recent delinquency behavior. Bill statement amounts measure balance pressure. Payment amounts proxy recent repayment effort or liquidity. Demographic attributes are used for diagnostic subgroup analysis and should not be interpreted as causal mechanisms.

The target is defined as $y_i = 1$ when borrower i defaults in the next month and $y_i = 0$

Table 1: Conceptual variable groups in the credit-risk dataset.

Feature group	Examples	Research interpretation
Credit capacity	Credit limit	Proxy for available credit and potential exposure.
Repayment history	Recent payment status variables	Captures delinquency patterns and payment stress.
Bill amounts	Monthly statement balances	Measures balance pressure and utilization behavior.
Payment amounts	Monthly repayments	Measures recent effort or capacity to reduce balances.
Demographics	Age and related attributes	Used for subgroup diagnostics, not causal claims.

otherwise. The modeling task is to estimate

$$p_i = \Pr(y_i = 1 \mid x_i), \quad (1)$$

where x_i denotes the borrower feature vector. This probability estimate is not itself the lending decision. It is an input into a policy rule that may approve, reject, price, or route a borrower for additional review.

5. Formal Notation

For clarity, Table 2 summarizes the principal notation used throughout the paper.

Table 2: Notation used in the validation framework.

Symbol	Meaning
y_i	Observed default outcome for borrower i , where 1 indicates default and 0 indicates non-default.
x_i	Feature vector for borrower i .
p_i	Predicted probability of default, $\Pr(y_i = 1 \mid x_i)$.
τ	Maximum default-risk threshold allowed by a lending policy.
$d_i(\tau)$	Approval decision under threshold τ .
E_i	Borrower exposure proxy.
r	Profit rate if an approved borrower repays.
λ	Loss-given-default rate.
$U_i(\tau)$	Borrower-level utility under threshold τ .
FAR	False approval rate among approved borrowers.
FRR	False rejection rate among true repayers.
ECE	Expected Calibration Error.
MCE	Maximum Calibration Error.

6. Preprocessing, Reproducibility, and Leakage Control

Preprocessing is designed to make the pipeline reproducible and defensible. Column names are standardized so that downstream analysis can rely on consistent variable names even when source metadata differ. Missing numeric values are imputed using the median, a conservative choice that is robust to extreme observations and preserves sample size. Numeric features are standardized for Logistic Regression so coefficient estimation is not dominated

by variables measured on larger scales. Tree-based models are less sensitive to scaling, but a unified pipeline improves comparability.

All transformations that learn from data are fit only on the training split. The test split is reserved for final evaluation. This design prevents data leakage, which would occur if information from the test set influenced imputation, scaling, feature construction, model fitting, threshold selection, or calibration analysis. Leakage is especially problematic in credit-risk modeling because it can make a model appear more stable, accurate, or profitable than it would be in deployment.

$$D = D_{\text{train}} \cup D_{\text{test}}, \quad D_{\text{train}} \cap D_{\text{test}} = \emptyset, \quad (2)$$

$$T = T(D_{\text{train}}), \quad X_{\text{train}}^* = T(X_{\text{train}}), \quad X_{\text{test}}^* = T(X_{\text{test}}). \quad (3)$$

The notation emphasizes that the test set is transformed using parameters learned exclusively from the training set. This principle is essential for credible validation. If test observations influence preprocessing, the evaluation becomes partially circular and overstates expected out-of-sample performance.

Data cleaning also removes pure identifier fields when they contain one unique value per row and do not represent meaningful borrower behavior. Identifier variables can encourage memorization of row structure rather than learning risk patterns. The modeling focus is therefore maintained on repayment history, balance pressure, payment behavior, credit exposure, and engineered risk signals.

The preprocessing choices are intentionally conservative. The objective is not to maximize a benchmark score through aggressive tuning, but to construct a transparent and reproducible validation design. Each preprocessing step supports one or more of four goals: reproducibility, leakage prevention, robustness, and interpretability.

7. Feature Engineering

Feature engineering translates raw account records into behavioral risk indicators. The dataset contains repeated monthly variables for bill amounts, repayment amounts, and repayment status. These variables are informative individually, but aggregated features can make borrower behavior more interpretable and more suitable for decision-oriented analysis.

Average bill amount summarizes balance pressure across observed months. A borrower with persistently high statement balances may be experiencing greater credit demand or liquidity pressure. The feature is defined as

$$\text{AverageBill}_i = \frac{1}{K} \sum_{k=1}^K \text{Bill}_{ik}. \quad (4)$$

The bill-to-limit ratio approximates utilization intensity by comparing average balances with available credit capacity:

$$\text{Utilization}_i = \frac{\text{AverageBill}_i}{\text{CreditLimit}_i + \epsilon}. \quad (5)$$

Higher utilization may be associated with higher credit stress, although the analysis treats it as a predictive signal rather than a causal claim.

A delayed-month count summarizes repayment stress. It counts the number of months in which the recorded payment status indicates delay:

$$\text{DelayCount}_i = \sum_{k=1}^K \mathbf{1}\{\text{PayStatus}_{ik} > 0\}. \quad (6)$$

Payment aggregates summarize recent repayment effort relative to outstanding balances. These engineered variables link model inputs to economically meaningful mechanisms: exposure, utilization, delinquency, and repayment behavior. They also support subgroup analysis because borrowers with prior payment delay or low credit limits can be evaluated separately.

A careful distinction is maintained between prediction and causality. A feature that improves predictive performance does not necessarily cause default. High utilization may predict default because it reflects liquidity stress, but this study does not claim that utilization mechanically causes default in isolation. The dataset is observational, and the purpose is model validation rather than causal identification.

8. Modeling Pipeline

The modeling pipeline contains five stages: data loading, preprocessing, feature construction, model fitting, and decision-oriented evaluation. Each model estimates a default probability for every borrower in the test set. These probabilities are then converted into decisions through a threshold rule:

$$d_i(\tau) = \mathbf{1}\{p_i < \tau\}. \quad (7)$$

Here, $d_i(\tau) = 1$ indicates approval and τ denotes the maximum default-risk threshold tolerated by the policy. A lower threshold is more conservative because fewer borrowers qualify for approval. A higher threshold expands approval volume but may increase default losses.

Logistic Regression is included as a transparent baseline. Random Forest is included because it captures nonlinear effects and interactions through many decision trees trained on resampled data. Gradient Boosting is included because it builds trees sequentially and can focus on observations that earlier trees model poorly. Together, the models represent a useful comparison between interpretable linear modeling and flexible nonlinear ensembles.

Table 3: Models and validation concerns.

Model	Role	Validation concern
Logistic Regression	Interpretable linear baseline	May underfit nonlinear borrower-risk patterns.
Random Forest	Bagged nonlinear ensemble	May require calibration and interpretability diagnostics.
Gradient Boosting	Sequential nonlinear ensemble	May be sensitive to tuning and distribution shift.

The comparison is not framed as a simple ranking exercise. A model that achieves the highest AUC may not generate the strongest expected utility. A model with strong baseline utility may deteriorate more sharply under stress. A model with acceptable average performance may still be unreliable for a specific subgroup. For that reason, the core contribution of the study is the multi-metric evaluation design.

9. Financial Utility Framework

Credit decisions involve asymmetric consequences. Approving a borrower who defaults creates economic loss, while rejecting a borrower who would have repaid creates opportunity cost and may reduce access to credit. The proposed framework evaluates model decisions through expected financial utility rather than classification labels alone.

For borrower i , let E_i denote exposure, r denote the profit rate if the borrower repays, and λ denote the loss-given-default rate. Borrower-level utility under threshold τ is defined

as

$$U_i(\tau) = d_i(\tau) [(1 - y_i)rE_i - y_i\lambda E_i]. \quad (8)$$

Average portfolio utility is then the mean of borrower-level utility across the evaluation sample:

$$U(\tau) = \frac{1}{n} \sum_{i=1}^n U_i(\tau). \quad (9)$$

This formulation links prediction to decision-making. Instead of asking only whether the model labels a borrower correctly, the framework asks what happens economically after acting on the model decision. This is the relevant operational question for lending policy.

The false approval rate is the share of approved borrowers who actually default. It captures the default risk embedded in the approved population:

$$\text{FAR}(\tau) = \frac{\sum_i \mathbf{1}\{d_i(\tau) = 1 \wedge y_i = 1\}}{\sum_i \mathbf{1}\{d_i(\tau) = 1\}}. \quad (10)$$

The false rejection rate is the share of true repayers who are rejected. It captures the opportunity cost of excessive conservatism:

$$\text{FRR}(\tau) = \frac{\sum_i \mathbf{1}\{d_i(\tau) = 0 \wedge y_i = 0\}}{\sum_i \mathbf{1}\{y_i = 0\}}. \quad (11)$$

A conservative threshold may reduce false approvals but increase false rejections. An aggressive threshold may expand approvals and potential profit but increase default loss. Threshold selection is therefore a policy choice rather than a purely technical parameter.

10. Calibration Analysis

Calibration evaluates whether predicted probabilities are numerically reliable. This differs from ranking performance. A model may rank borrowers correctly while still assigning probabilities that are too high or too low. In lending, this matters because thresholds, pricing, reserve planning, and stress monitoring often depend on probability values.

The Brier score measures mean squared probability error and therefore combines aspects of calibration and discrimination:

$$\text{BrierScore} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (12)$$

To evaluate calibration by probability intervals, the test set is partitioned into M bins according to predicted default probability. Expected Calibration Error summarizes the weighted average difference between observed default frequency and average predicted probability across bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{ObservedDefaultRate}(B_m) - \text{AveragePrediction}(B_m)|. \quad (13)$$

Maximum Calibration Error captures the largest bin-level deviation between observed default frequency and predicted probability:

$$\text{MCE} = \max_m |\text{ObservedDefaultRate}(B_m) - \text{AveragePrediction}(B_m)|. \quad (14)$$

Calibration analysis is essential because the model output is a probability rather than merely a label. If probabilities are too low, the lender may approve too many risky borrowers. If probabilities are too high, the lender may reject too many borrowers who would have repaid. Calibration therefore connects statistical learning directly to risk governance.

11. Stress Testing Design

Stress testing evaluates how model behavior changes when borrower conditions deteriorate. Because the dataset is cross-sectional rather than a macroeconomic panel, the stress design is interpreted as a controlled simulation rather than a forecast of a specific recession. The purpose is to measure relative model sensitivity under plausible deterioration in risk-relevant features.

A generic stress transformation can be represented as a mapping that modifies selected borrower features according to stress intensity δ :

$$x_i^{\text{stress}} = g(x_i; \delta). \quad (15)$$

The analysis focuses on economically meaningful stress channels: repayment-delay severity and utilization pressure. Increasing payment delay represents worsening ability to keep accounts current. Increasing utilization represents higher balance pressure relative to available credit. After the stress transformation, each model produces revised default probabilities:

$$p_i^{\text{stress}} = f(x_i^{\text{stress}}). \quad (16)$$

The stress effect on utility is the difference between stressed and baseline utility:

$$\Delta U = U_{\text{stress}} - U_{\text{baseline}}. \quad (17)$$

The important question is not whether utility declines under stress; it normally should. The more informative question is whether utility declines gradually or collapses sharply, and whether some models are more sensitive than others. A robust model should not become unstable under moderate distributional perturbation.

Stress testing also supports subgroup diagnostics. Some borrower groups may be more sensitive to utilization or repayment-delay changes. If stress causes subgroup utility gaps or calibration gaps to widen, a model may create concentrated decision risk even when its overall portfolio average remains acceptable.

12. Subgroup Robustness and Bias Diagnostics

Subgroup analysis evaluates whether model decisions and errors are distributed consistently across meaningful borrower groups. The subgroups are derived from available variables and interpreted in credit-risk terms: young borrowers, older borrowers, low-credit-limit borrowers, and borrowers with prior payment delays. The analysis is diagnostic rather than legal or causal. Its purpose is to identify where model utility, calibration, approval behavior, or stress sensitivity may be unstable.

Table 4: Subgroup definitions for robustness diagnostics.

Subgroup	Definition	Reason for inclusion
Young borrowers	Age at or below 30	Tests early-credit-life behavior and calibration.
Older borrowers	Age at or above 55	Tests whether model behavior differs for older applicants.
Low credit limit	Limit below sample median	Proxy for lower credit capacity or smaller exposure.
Prior payment delay	Two or more delayed months	Direct signal of payment stress and default risk.

For each subgroup G , the same decision metrics used in the overall evaluation are re-computed. This allows the analysis to determine whether utility, false approvals, false rejections, and calibration errors differ across borrower segments:

$$U_G(\tau) = \frac{1}{|G|} \sum_{i \in G} U_i(\tau). \quad (18)$$

Subgroup utility degradation under stress is defined as the difference between stressed and baseline subgroup utility:

$$\Delta U_G = U_G^{\text{stress}} - U_G^{\text{baseline}}. \quad (19)$$

This design helps separate possible explanations for poor subgroup performance. A subgroup may have lower utility because its observed default rate is higher, because probabilities are miscalibrated for that group, because the threshold creates a harmful tradeoff, or because stress affects that group more strongly. Distinguishing among these explanations is necessary for credible research interpretation.

13. Bootstrap Uncertainty and Paired Significance Testing

Point estimates should not be interpreted as fixed truths. Average utility, loss, calibration error, and model differences are estimated from finite data. Bootstrap resampling provides a practical empirical method for quantifying uncertainty around these estimates.

In each bootstrap iteration, the test set is sampled with replacement to create a replicate sample of the same size as the original test set. Metrics are recomputed on each replicate. The resulting distribution approximates the sampling variability of the metric under the observed data-generating environment:

$$\text{CI}_{95}(U) = \left[q_{0.025}(\{U^{(b)}\}), q_{0.975}(\{U^{(b)}\}) \right]. \quad (20)$$

The study also uses paired bootstrap comparison. Instead of evaluating model A and model B on unrelated samples, both models are evaluated on the same bootstrap replicate. The difference in utility is computed within each replicate:

$$\Delta_{A,B}^{(b)} = U_A^{(b)} - U_B^{(b)}. \quad (21)$$

If the confidence interval for the paired difference excludes zero, the result suggests that one model has a statistically meaningful utility advantage under the selected assumptions. This conclusion does not establish universal superiority because it remains conditional on the dataset, financial parameters, threshold policy, and stress design. However, it is more rigorous than claiming superiority from a single point estimate.

14. Experimental Workflow

The full experiment follows a reproducible workflow. First, the dataset is loaded and standardized. Second, engineered features are created. Third, the data are split into training and test sets. Fourth, each model is trained using the same target definition and comparable preprocessing. Fifth, test-set probabilities are generated. Sixth, threshold rules convert probabilities into approval or rejection decisions. Seventh, predictive, financial, calibration, stress, subgroup, and bootstrap metrics are computed. Finally, results are exported for reporting.

The workflow is modular. A researcher can replace the dataset, add model classes, modify utility assumptions, or define alternative stress scenarios without rewriting the entire

Table 5: Experimental workflow and outputs.

Stage	Output	Purpose
Data loading	Clean borrower table	Creates reproducible input data.
Feature engineering	Behavioral risk variables	Summarizes utilization and repayment stress.
Model fitting	LR, RF, and GB estimators	Compares linear and nonlinear approaches.
Probability prediction	Default probabilities	Supports thresholding and calibration analysis.
Decision evaluation	Utility, FAR, and FRR	Connects model output to lending outcomes.
Stress and subgroup tests	Robustness metrics	Examines distribution shift and group stability.
Bootstrap	Confidence intervals	Quantifies uncertainty and model differences.

pipeline. This modularity supports reproducibility and auditability because each reported metric can be traced back to a probability estimate, threshold decision, and observed outcome.

15. Results Interpretation Framework

Because exact numerical values depend on the final computational run, threshold selection, and utility parameters, the most defensible interpretation should rely on a dashboard of evidence rather than unsupported point estimates. The first results table should report predictive metrics such as accuracy, precision, recall, F1, and AUC. The second should report decision metrics such as approval rate, false approval rate, false rejection rate, expected loss, and average utility. The third should report calibration metrics including Brier score, ECE, and MCE. The fourth should report subgroup metrics. The fifth should report bootstrap intervals and paired utility differences.

A rigorous model-selection statement would not merely say that one algorithm is best. A stronger conclusion would state that the selected model offers the most favorable utility-risk tradeoff under the specified threshold and financial assumptions while maintaining acceptable calibration and showing less stress-related degradation than competing models. Such a statement is more scientifically defensible because it is conditional on explicit assumptions and multiple validation criteria.

The framework also highlights tradeoffs. Logistic Regression may be more transparent and stable but less flexible. Random Forest may capture nonlinear borrower behavior but require calibration assessment. Gradient Boosting may provide strong predictive capacity but can be sensitive to tuning and distribution shift. The preferred model is therefore not automatically the one with the highest AUC; it is the model with the strongest combined profile across utility, calibration, robustness, interpretability, and uncertainty.

16. Discussion

The principal contribution of this study is the integration of predictive modeling, decision-theoretic evaluation, calibration assessment, distributional robustness analysis, subgroup diagnostics, and statistical uncertainty quantification within a unified consumer credit-risk validation framework. Many benchmark studies emphasize predictive discrimination metrics, but such metrics alone do not adequately characterize operational performance in

lending environments where probability estimates are converted into approval policies and financial outcomes.

The decision-risk framing is especially important because deployment is threshold-based. A predicted probability is not the final product. The final product is a decision rule that approves, rejects, prices, or escalates borrowers. Therefore, the appropriate unit of evaluation is not only the prediction but also the consequence of acting on that prediction.

The calibration component strengthens the research design because probability quality is essential for financial decision-making. A lender cannot reliably price, threshold, or reserve against default risk if predicted probabilities are miscalibrated. By including Brier score, ECE, MCE, and reliability analysis, the framework evaluates probability reliability rather than label prediction alone.

The stress-testing component connects machine learning with financial risk management. In stable conditions, many models may appear acceptable. Under stress, model differences may become more visible. Some models may overreact to perturbed variables, while others may remain more stable. This analysis raises questions about distribution shift, model monitoring, and the limits of public benchmark datasets.

The subgroup component shows that average performance can conceal important heterogeneity. If a model performs well overall but poorly for borrowers with low credit limits or prior delays, it may create concentrated decision risk. Subgroup diagnostics can identify where recalibration, threshold adjustment, additional review, or fairness-aware constraints may be necessary.

17. Methodological and Practical Contributions

This study makes a methodological contribution rather than a claim that one public-dataset model should be deployed in practice. The contribution is the validation framework. The framework demonstrates how a standard supervised-learning credit-risk analysis can be extended into a decision-aware research design by connecting model outputs to decisions, financial consequences, calibration quality, stress sensitivity, subgroup behavior, and statistical uncertainty.

This contribution matters because many practical research questions begin where ordinary performance reporting ends. If two models have similar AUC, which should be preferred under a given financial objective? If a model has high average utility but poor calibration, should its probabilities be trusted for pricing or thresholding? If a model performs well overall but degrades for low-limit borrowers under stress, is the average result sufficient? If a paired bootstrap interval crosses zero, should the apparent improvement be treated as meaningful? These questions require statistical reasoning and explicit decision assumptions.

The framework also reflects responsible modeling. Credit decisions affect both borrowers and institutions. Even when the dataset is used for methodological research, evaluation should reflect the consequences of model decisions. Subgroup analysis is therefore not treated as a superficial fairness check; it is integrated into the decision-risk framework through utility, false approvals, false rejections, calibration gaps, and stress degradation.

18. Threats to Validity

Several threats to validity should be recognized. Internal validity may be affected by pre-processing choices, threshold definitions, or the specific construction of stress variables. External validity is limited because the dataset is public and historical, not a current production credit portfolio. Construct validity depends on whether the chosen financial utility

assumptions adequately approximate real lending outcomes. Statistical conclusion validity depends on test-set size, bootstrap design, and model-selection discipline.

The stress scenarios are intentionally stylized. They are designed to examine sensitivity, not to reproduce a complete macroeconomic shock. Similarly, subgroup definitions are diagnostic categories derived from available variables, not causal claims about borrower identity. These limitations do not invalidate the framework, but they require conclusions to be stated conditionally and carefully.

19. Limitations

This study has several limitations. First, the UCI dataset is public and historically useful, but it is not a current production credit portfolio. Results should therefore be interpreted as a methodological demonstration rather than a direct deployment recommendation. Second, the stress scenarios are simulated through feature perturbations. They do not fully capture macroeconomic dynamics such as unemployment shocks, interest-rate changes, inflation, policy interventions, or borrower income variation.

Third, the utility framework depends on assumed profit and loss parameters. Different institutions may use different profit margins, loss-given-default assumptions, capital requirements, regulatory constraints, and portfolio objectives. The framework is designed to be adjustable, but model preference may change when assumptions change. This conditionality is both a strength and a limitation: it makes the framework flexible, but it also means that conclusions must be stated carefully.

Fourth, subgroup analysis is diagnostic rather than causal. Differences across groups may reflect many interacting factors. The study does not claim that age or credit limit causes default. It uses subgroup definitions to examine model behavior and identify possible instability. A causal analysis would require a different design, stronger assumptions, and potentially external data.

Fifth, the models examined here are standard supervised-learning models. More advanced approaches, including calibrated gradient boosting, survival analysis, reject inference, causal modeling, conformal prediction, and distributionally robust optimization, could extend the framework. The chosen model set is intentionally understandable while still covering both linear and nonlinear families.

20. Future Research

Several extensions would strengthen the research design. First, temporal validation could evaluate whether a model trained on one period performs well on later observations. This would be more realistic than a random train-test split because credit-risk systems are deployed over time. Second, external validation on another credit dataset would test generalization and reduce dependence on one public benchmark.

Third, threshold optimization could be formulated as a constrained policy problem that balances expected utility, approval volume, calibration, and subgroup constraints. Fourth, calibration under stress could be examined directly. A model may remain useful if monitored and recalibrated, but become risky if treated as static after distribution shift.

Uncertainty-aware decision rules are another promising extension. Instead of making a hard approval or rejection decision for every borrower, a model could route borderline cases to manual review. This would connect bootstrap uncertainty, prediction uncertainty, and operational policy design. In a real lending environment, the model need not replace human review; it can prioritize and structure it.

Finally, explainability could be added through feature importance, partial dependence,

SHAP-style summaries, or counterfactual analysis. These methods could help explain why models approve or reject borrowers and whether specific variables drive subgroup differences. Explainability would make the framework stronger for governance, auditability, and stakeholder communication.

21. Research Implications

The theoretical implication is that credit-risk validation should not be reduced to discrimination metrics. Probability estimates become meaningful only when they are attached to an operating policy and an economic interpretation. The methodological implication is that model evaluation should report calibration, utility, subgroup behavior, stress degradation, and uncertainty in a unified evidence set. The practical implication is that financial institutions should monitor decision quality, not merely model accuracy, because the costs of errors are asymmetric and may change under stress.

For PhD-level research presentation, the key takeaway is that the project is not simply a credit-default classifier. It is a structured validation framework that asks whether a model remains credible when its predictions are transformed into lending decisions under uncertainty, changing borrower conditions, and subgroup heterogeneity.

22. Conclusion

This study develops a decision-aware validation framework for consumer credit models under distributional stress. Using the UCI Default of Credit Card Clients dataset, the analysis evaluates Logistic Regression, Random Forest, and Gradient Boosting across predictive performance, financial utility, calibration, stress robustness, subgroup behavior, and statistical uncertainty. The central argument is that credit model evaluation should be decision-aware. A model should not be judged only by how accurately it predicts default, but also by how its probabilities behave when converted into lending decisions under uncertainty.

The framework reframes consumer credit modeling from a narrow classification task into a broader decision-risk problem. This reframing is the strongest contribution of the study. It demonstrates that model validity in credit risk depends on probability quality, threshold design, asymmetric financial consequences, subgroup heterogeneity, stress sensitivity, and uncertainty. Future research can extend the framework through temporal validation, external datasets, fairness constraints, uncertainty-aware thresholds, explainability, and real-world monitoring.

23. Reproducibility Statement

The analysis is designed to be reproducible from raw data ingestion through final metric reporting. The workflow separates data loading, preprocessing, feature construction, model estimation, probability prediction, threshold evaluation, stress testing, subgroup diagnostics, and bootstrap inference into independent modules. This structure allows each result to be traced back to a specific modeling assumption and a specific decision rule. It also permits alternative assumptions to be tested without changing the conceptual framework.

For research presentation, reproducibility is not limited to sharing code. It also requires explaining why each metric is included, how each threshold is selected, and how uncertainty is quantified. The proposed design therefore treats documentation as part of model validation. A reader should be able to understand not only what the model predicted, but how those predictions were transformed into decisions and how sensitive the conclusions are to stress and sampling variability.

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